

EXPRESSIONS, FUNCTIONS, AND CONTROL Solutions

COMPUTER SCIENCE MENTORS 61A

August 31, 2026 – September 4, 2026

1 Expressions

1. What Would Python Display?

```
>>> 3

3

>>> "cs61a"

'cs61a'

>>> x = 3
>>> x

3

>>> x = print("cs61a")
cs61a
>>> x

>>> print(print(print("cs61a")))

cs61a
None
None

>>> def f1(x):
...     return x + 1
>>> f1(3)

4

>>> f1(2) + f1(2 + 3)

9
```

```

>>> def f2(y):
...     return y / 0
>>> f2(4)

ZeroDivisionError: division by zero

>>> def f3(x, y):
...     if x > y:
...         return x
...     elif x == y:
...         return x + y
...     else:
...         return y
>>> f3(1, 2)

2
>>> f3(5, 5)

10
>>> 1 or 2 or 3

1
>>> 1 or 0 or 3

1
>>> 4 and (2 or 1/0)

2
>>> 0 or (not 1 and 3)

False
>>> (2 or 1/0) and (False or (True and (0 or 1)))

1

```

2 Functions

1. For the following expressions, simplify the operands in the order of evaluation of the entire expression

Example: `add(3, mul(4, 5))`

Order of Evaluation: `add(3, mul(4, 5))` $\rightarrow$ `add(3, 20)` $\rightarrow$ 23

(a) `add(1, mul(2, 3))`

```

add(1, mul(2, 3))
add(1, 6)
7

```

(b) `add(mul(2, 3), add(1, 4))`

```
add(mul(2, 3), add(1, 4))
add(6, add(1, 4))
add(6, 5)
11
```

(c) `max(mul(1, 2), add(5, 6), 3, mul(mul(3, 4), 1), 7)`

```
max(mul(1, 2), add(5, 6), 3, mul(mul(3, 4), 1), 7)
max(2, add(5, 6), 3, mul(mul(3, 4), 1), 7)
max(2, 11, 3, mul(mul(3, 4), 1), 7)
max(2, 11, 3, mul(12, 1), 7)
max(2, 11, 3, 12, 7)
12
```

2. Sally really wants to know what she needs to score to pass her class! Fill out the function `score_needed` that takes in *current_grade*, her current grade, *num_exams*, the number of exams taken so far (not including her final exam), and *pass_grade*, the grade needed to pass. Assume that every exam, including the final exam, has the same weight.

```
def score_needed(current_grade, num_exams, pass_grade):
    '''
    >>> score_needed(89, 4, 90)
    94
    >>> score_needed(65, 2, 70)
    80
    >>> score_needed(77, 10, 78)
    88
    '''
```

```
def score_needed(current_grade, num_exams, pass_grade):
    total_pts = current_grade * num_exams
    total_needed = pass_grade * (num_exams + 1)
    return total_needed - total_pts
```

3 Environment Diagrams

1. Give the environment diagram and console output that result from running the following code.

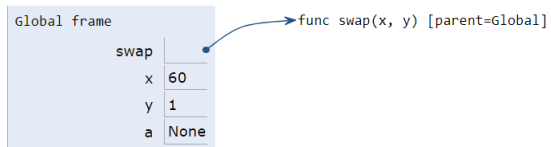
```
def swap(x, y):  
    x, y = y, x  
    return print("Swapped!", x, y)
```

```
x, y = 60, 1  
a = swap(x, y)  
swap(a, y)
```

Print output (drag lower right corner to resize)

```
Swapped! 1 60  
Swapped! 1 None
```

Frames Objects



f1: swap [parent=Global]

x	1
y	60
Return value	None

f2: swap [parent=Global]

x	1
y	None
Return value	None

<https://tinyurl.com/y68m6qdj>

1. Write a function that returns `True` if a number is divisible by 4, 1 if a number is divisible by 7 and is not already divisible by 4, and returns `False` if neither condition is fulfilled.

```
def divisibility_check(num):
```

```
def divisibility_check(num):  
    if num % 4 == 0:  
        return True  
    elif num % 7 == 0:  
        return 1  
    else:  
        return False
```

This also works as an alternate solution:

```
def divisibility_check(num):  
    return True if num % 4 == 0 else 1 if num % 7 == 0 else False
```

2. Implement `fizzbuzz(n)`, which prints numbers from 1 to `n` (inclusive). However, for numbers divisible by 3, print “fizz”. For numbers divisible by 5, print “buzz”. For numbers divisible by both 3 and 5, print “fizzbuzz”.

```
def fizzbuzz(n):  
    """  
    >>> result = fizzbuzz(16)  
    1  
    2  
    fizz  
    4  
    buzz  
    fizz  
    7  
    8  
    fizz  
    buzz  
    11  
    fizz  
    13  
    14  
    fizzbuzz  
    16  
    >>> result is None  
    True  
    """  
  
    i = 1  
    while i <= n:  
        if i % 3 == 0 and i % 5 == 0:  
            print('fizzbuzz')  
        elif i % 3 == 0:  
            print('fizz')  
        elif i % 5 == 0:  
            print('buzz')  
        else:  
            print(i)  
        i += 1
```

We must put the condition `i % 3 == 0 and i % 5 == 0` in the first `if`. For example, if we were to write the body of the while loop with the first two conditions switched.

```
if i % 3 == 0:  
    print('fizz')  
elif i % 3 == 0 and i % 5 == 0:  
    print('fizzbuzz')  
elif i % 5 == 0:  
    print('buzz')  
else:  
    print(i)
```

then we may print out ‘fizz’ for a number like 15 which would be incorrect.

5 Higher-Order Functions

1. Write a function, `whole_sum`, which takes in an integer, `n`. It returns another function which takes in an integer, and returns `True` if the digits of that integer sum to `n` and `False` otherwise.

```
def whole_sum(n):  
    """  
    >>> whole_sum(21) (777)  
    True  
    >>> whole_sum(142) (10010101010)  
    False  
    """  
    def check(x):  
  
        _____  
  
        while _____:  
  
            last = _____  
  
            _____  
  
            _____  
  
        return _____  
  
    return _____
```

```
def whole_sum(n):  
    def check(x):  
        total = 0  
        while x > 0:  
            last = x % 10  
            x = x // 10  
            total += last  
        return total == n  
    return check
```